



OpenVVF

The open-source, safety-engineered power conversion platform — 50 kW to 1 MW+.

400 V class - 465 A

Continuous · 600 A peak (60 s) ·
150/200/400 V by DC-link selection

Dual-MCU safety

FOC controller + independent safety
coprocessor

100% open

Hardware, firmware, safety analyses &
live telemetry

SMART POWER LAB · UC SANTA CRUZ

BUILT + BENCH-TESTED

Proven on real hardware

Chassis Size 2 (~100 kW class) has run a **180 V bring-up**, a **1-hour endurance run**, a **20-minute reversal stress test**, and motor self-commissioning — **10 formal test reports**. Next: a Size 2 chassis for our **2018 Zero DSR** electric motorcycle.

PLATFORM FAMILY

SAFETY-ENGINEERED

Designed like it matters

Dual-MCU architecture with **six redundant safe-state pathways**; HARA + FMEA with **ISO 26262** and **IEC 61800-5-2** mapping; every hazard traces to a safety requirement and a fault-injection test.

OPEN EVERYTHING

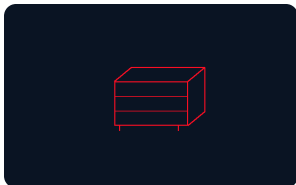
Nothing sealed, nothing hidden

Schematics, firmware, the **RTE Studio** model-based workflow (no MATLAB required; **ADRC & MPCC benchmarked against FOC**), safety analyses, and **interactive telemetry logs** — all public at docs.openvvvf.org.

One control module — four power stages

CONTROL MODULE Dual-MCU control module

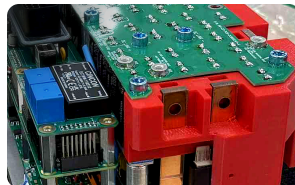
FOC motor control + independent safety coprocessor · USB-B / dual CAN FD / expansion · RTE Studio + Telemetry Viewer



Size 1

~50 kW class inverter

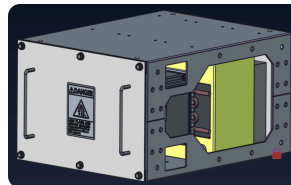
● EARLY DESIGN



Size 2 Inverter

~100 kW · 400 V / 465 A

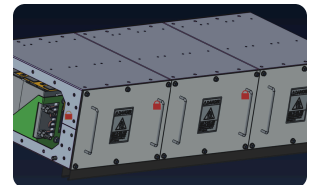
● IN VALIDATION



Size 2 DC-DC

Charging & microgrids

● IN PROGRESS



Size 3

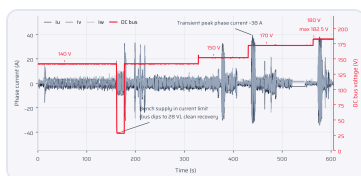
MW class · 900 V / 1400 A

● DESIGN IN PROGRESS

Size 2 power section

Three IGBT half-bridges conduction-cooled to an aluminum baseplate — ultra-low-inductance DC-link

IGBT TODAY · SIC-READY



Live telemetry, not PDF promises

Every formal test report links its interactive data log — viewable client-side in the browser-based Telemetry Viewer, no upload required. Scan to open the 180 V bring-up: bus stepped 140 to 180 V, ~38 A peak, clean recovery from a bench-supply current-limit event.



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